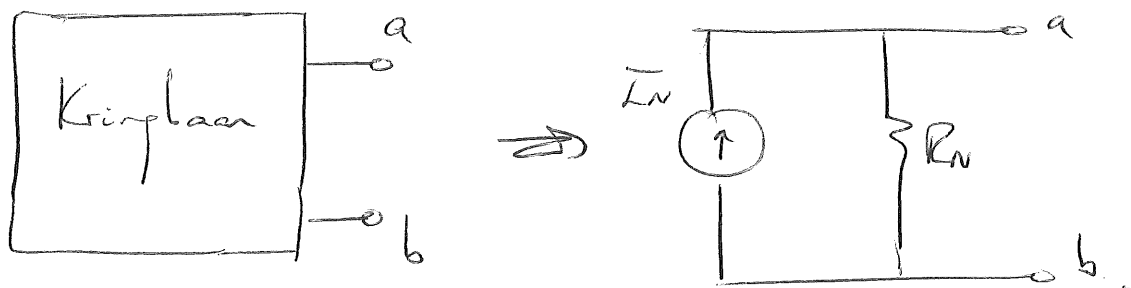


Norton se Teorie

→ Norton se teorie

- * Veroang enige 2-terminaal kringbaan met 'n stroombaan in parallel met 'n resistor sodat die twee kringbane ekwivalent is.



→ I_N

- * Norton stroom is stroom tussen terminale a en b indien kortsluiting tussen terminale gevorm word.
(Kortsluitstroom!).

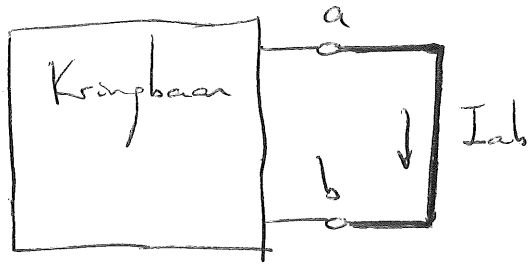
→ R_N

- * Ekwivalente weerstand indien alle onafh bronne afgeskakel is.
(Dieselfde as R_{th} .)

* Hoe om I_N en R_N te berekenen.

(2)

I_N



* $I_{sc} \Rightarrow$ 'short-circuit' stroom

* $I_{ab} = I_{sc} = I_N$

* I_N is stroom van a na b indien kortsluiting bygevoegd word.

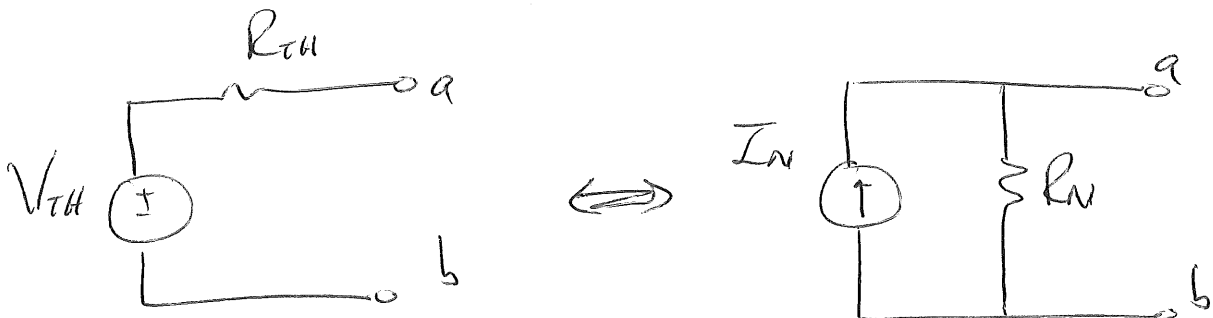
→ Bereken I_{ab} deur:

- * KVL; KCL; Σ -Wet
- * Lusanalyse; Nodeanalyse
- * Superposisie; Brontransf.

R_N

Norton weerstand en Thevenin weerstand is dieselfde $R_N = R_{TH}$.

Onthou brontransformasie:

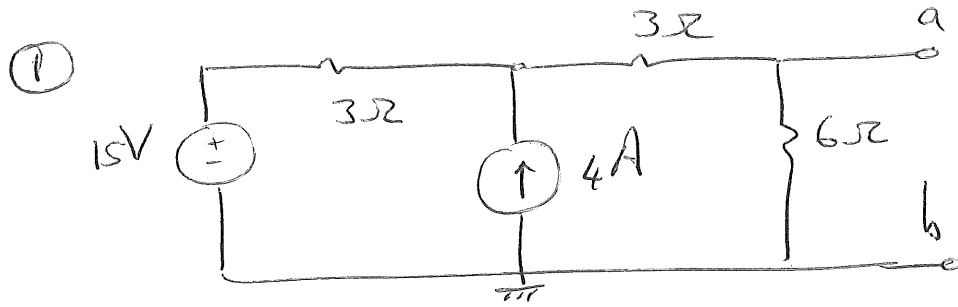


$$V_{TH} = I_N \cdot R_N$$

$$\& V_{TH} = I_N \cdot V_{TH}$$

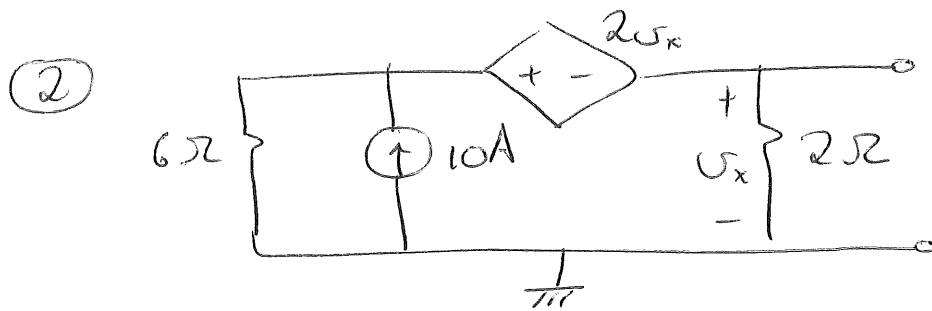
Voorbeelden:

3



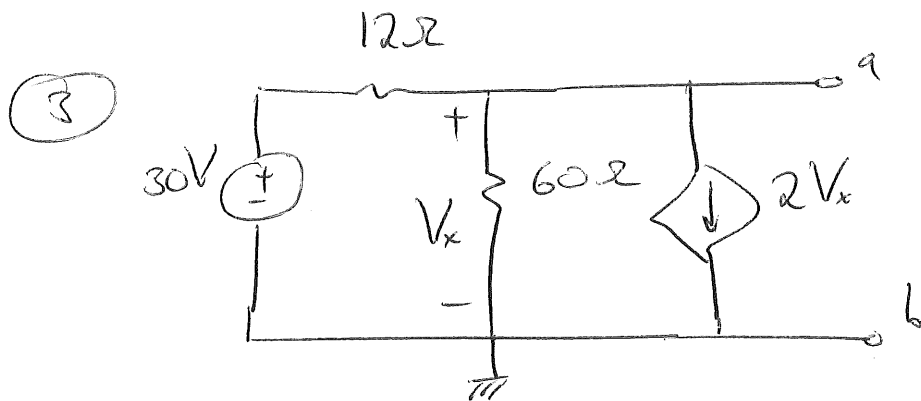
$$I_N = ?$$

$$R_N = ?$$



$$I_N = ?$$

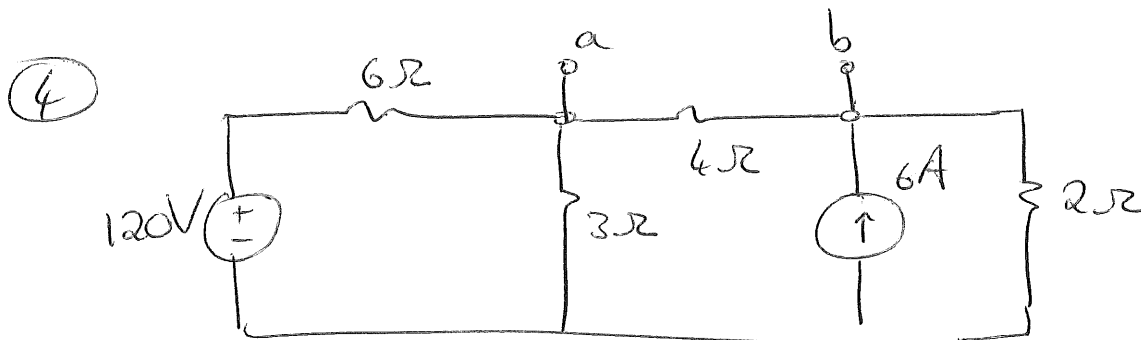
$$R_N = ?$$



$$V_{TH} = ?$$

$$I_N = ?$$

$$R_N = R_{TH} = ?$$



$$V_{TH} = ?$$

$$I_N = ?$$

$$R_N = ?$$